attending a fair where 800 people tried to guess the weight of a dead ox. After the prize was awarded, Galton asked if he could see the guesses. Most of them were way off the mark. Galton was shocked, however, to see that the average guess was 1,197 pounds, while the actual weight of the ox was 1,196 pounds! There was a true signal hidden among all the noise. Averaging was able to make sense from apparent randomness.

Absolute momentum reduces noise by looking at two reference points in time. It essentially asks if today's price is higher or lower than the price was 12 months ago. Moving averages reduce noise by smoothing it out through averaging, just as Galton had done.

MARKET TIMING USING VALUATION

Some practitioners believe they can time the market by paying attention to valuation metrics, such as the Shiller 10-year Cyclically Adjusted Price Earnings (CAPE) ratio. CAPE is calculated by taking the S&P 500 and dividing it by the average of 10 years' worth of earnings. The current CAPE ratio is then compared to its long-term average of about 17. Investors look for mean reversion back toward this average. Historically, CAPE ratios under 10 have led to future annual stock market returns of over 20%, while CAPE ratios over 20 have given future annual stock market returns of only 5%.

The current CAPE level of about 26 indicates that the U.S. stock market is expensive compared to historic norms. However, this does not necessarily mean that stocks are near a bull market top. Those who got out of the market in 1996, when the CAPE was at approximately the same level as it is now, would have missed out on seeing the U.S. market more than double in value as the CAPE ratio rose to over 40 in 1999–2000. Paul Tudor Jones noted that the last one-third of a bull market is often the most dramatic, where mania runs wild and prices go parabolic. Market timing based on valuation could miss out on all this.

In addition, the normal level of corporate earnings may have changed over time, and studies of past CAPE levels may have overfit the data. Valuation metrics are only capable of giving a crude approximation of what future returns might be like.[7]